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RESEARCH ARTICLE

Optimized Fourth-Order Symmetric S-Curve Trajectory for Vibration Control in Dynamic Systems

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ABSTRACT In a high-speed and high-accuracy point-to-point (P-T-P) motion application, the induction of structural vibration is a common issue. This paper introduces a novel critical motion parameter (CMP) method to appropriately tune the 15-segment fourth-order symmetrical (15-SFOS) S-curve profile for achieving low vibration P-T-P motion in a flexible system. The method provides a tuning rule to optimize the 15-SFOS S-curve profile for suppressing induced residual vibration without explicitly addressing system dynamics, considering a given traveling distance, traveling time (TT), and actuator capacities. The method is formulated with a unitization approach that treats time and displacement as one unit each. This approach is used to present the characteristics of the 15-SFOS S-curve profile as nomograms. These nomograms are employed to determine motion parameter (MP) magnitudes without the need for calculations, according to the solution provided by the CMP method. The proposed trajectory is expressed in both the generalized polynomial equation and the time-Laplace domain method. The feasibility of the proposed method is experimentally demonstrated on the setup of a flexible rotating link. The experimental results are also validated through simulated results.

INDEX TERMS Critical motion parameter, multiplication factor, residual vibration, S-curve profile, traveling time, unitized magnitude.

I. INTRODUCTION

P-T-P motion is widely used in high-precision applications, including robotic manipulators, computer numerical control (CNC) machines, flexible systems such as wafer-handling robots, and various actuator-driven mechanisms. The P-T-P motion requires trajectory planning so that an input motion can provide the desired performance. Trajectory planning is commonly implemented using control algorithms derived from predefined motion profiles, a technique widely adopted across various industries. Trapezoidal velocity profiles, also known as second-order profiles, are recognized for enabling rapid motion execution [1]. However, a major limitation of the trapezoidal velocity profile is the abrupt change in acceleration, which results in high jerk and consequently induces

vibrations [2]. To address these issues, S-curve motion profiles have been introduced and extensively studied [3], [4], [5]. The S-curve profile offers several advantages over the trapezoidal velocity profile, including smoother motion trajectories, reduced discontinuities in control signals, lower risk of drive overheating, and diminished vibration levels [6], [7], [8], [9], [10], [11].

To achieve vibration-free motion and improve actuator performance, the MP, such as velocity, acceleration, and jerk, of an S-curve profile must be optimized based on the dynamic characteristics of the system. In recent years, plenty of work has been conducted on optimizing an S-curve model. Meckl et al. presented an input-shaping technique for generating the input-forcing function to optimize the ramp-up and ramp-down times of the third-order S-curve profile [12]. This technique is tested on the undamped flexible system, and it is found that the optimum third-order profile reduces

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residual vibration more effectively compared to the second-order profile. Richard uses a similar approach for optimizing a third-order S-curve profile to suppress the induced residual vibration of an underdamped flexible system [13]. In another work presented by Cho and Park, the input shaping method is applied to linearly time-varying multi-body flexible systems using timely varying impulse sequences to reduce the vibration [14]. Singh and Vadali presented a bang-bang control filter as a time-optimal P-T-P trajectory [15]. The transfer function of this filter is designed to place multiple zeros to cancel all the poles of the vibratory system. The effectiveness of this profile is demonstrated on a simplified flexible system, such as the spring-mass system. In addition to free-time formulations, fixed-settling-time optimal control strategies are investigated to design control inputs that effectively reduce vibrations in flexible systems [16]. The analysis presented in reference [16] demonstrates that, under reasonable assumptions, both types of problems yield the same solution and are therefore equivalent.

Piazzini and Visioli proposed a system inversion-based method in which the desired output is predefined using a polynomial function, and the corresponding actuator input is derived through dynamic inversion [17]. Sahinkaya adopted a similar approach, replacing the polynomial output with a third-order exponential function to ensure smooth convergence to the target position [18]. An S-curve switching method based on a fuzzy system is proposed to estimate the optimal maximum velocity for a third-order S-curve profile [19]. This approach enables precise position control of a DC motor while mitigating vibrations induced by uncertain load conditions. In another study, a third-order asymmetric S-curve profile is proposed, incorporating a jerk ratio parameter [20], [21], [22] and a jerk period ratio parameter [23], [24] to effectively reduce overshoot and mitigate induced vibrations in flexible systems. Simultaneously, Bearee and Olabi also proposed an asymmetrical third-order S-curve profile based on the methodology of Finite Impulse response (FIR) filtering to reduce the time-varying response of residual vibration [25]. Ha et al. present a third-order asymmetrical S-curve profile for long-distance motions, utilizing a Laplace-domain approach to suppress residual vibrations in lightly damped systems [26]. Biagiotti and Melchiorri introduced FIR filters, referred to as rectangular smoothers, for designing time-optimal third-order S-curve polynomial trajectories [27]. In this work, the vibration problem is solved by appropriately shaping the frequency spectrum of motion. Simultaneously, embedded prefilters are applied to polynomial and trigonometric-based S-curve profiles to further attenuate the system vibrations. These prefilters are designed and characterized in the Laplace domain [28]. In a similar approach within the Laplace transform domain, the eigenvalue method is employed to derive an analytical solution for the field distribution as a function of displacement in the semiconducting material [29], [30], [31]. In another work, displacement in the elastic solids with a spherical

or cylindrical cavity is numerically determined using Finite Element Method [32], [33]. In [34], the effect of the fractional time derivative on the displacement of both solid and fluid phases in elastic media is studied using FEM. A similar method is adopted in [35] in which a solution is provided to address the component of displacement in a thermoelastic half-space with voids. An energy-based approach is presented for designing an asymmetric time-optimal third-order S-curve profile [36]. This approach effectively optimizes both the magnitude and phase terms in vibration, thereby optimizing the traveling and settling time of the flexible motion system. Li et al. presented a low vibration S-curve profile based on trigonometric function under the kinematic limitations of velocity, acceleration, and jerk to reduce transient residual vibration for high speed linear motion stages [37], [38]. Akdag and Sen conducted a comparative analysis to evaluate the effectiveness of the trapezoidal velocity profile and the third-order S-curve profile in reducing vibrations in a single-link flexible rotating manipulator [39]. As expected, the third-order S-curve profile was found to be significantly more effective than the trapezoidal velocity profile in reducing vibrations.

For high-precision operations, motion smoothness is a critical criterion, as it minimizes mechanical disturbances and improves controller performance. Consequently, higher-order trajectories (fourth-order and above) are recently developed to further improve actuator performance and enhance vibration reduction capabilities [40]. In the work presented by Fang et al., a sigmoid function-based S-curve profile is designed under the kinematic limitations of velocity, acceleration, jerk, and snap [41]. The profile is designed to reduce the structural residual vibration induced in a robotic arm. One study investigates trapezoidal velocity, third-order S-curve, and fourth-order S-curve profiles in conjunction with feedforward controllers, using an algorithm specifically designed to compute time-optimal trajectories [42]. Simultaneously, Fan et al. investigate the characteristics of the snap acceleration/deceleration (fourth-order) profile and present a time-optimal trajectory for transitioning between velocities under specified kinematic constraints on velocity, acceleration, jerk, and snap, with the objective of minimizing structural vibrations [43]. The studies in [41], [42], and [43] present a time-optimal algorithm that determines the minimum time based on kinematic constraints, without accounting for the system dynamics. These time-optimal algorithms can ensure the minimum time of the P-T-P motion, but they do not guarantee minimum vibration characteristics. Therefore, Lee and Ha proposed second-order, third-order S-curve, and fourth-order S-curve profiles to enable rapid motion with minimal vibration in flexible systems, considering various constraints on actuator capabilities [44]. In another recent work, a time-optimal S-curve trajectory under kinematic constraints, such as maximum velocity, maximum acceleration, and frequency constraints, is designed using FIR filters to suppress the vibration caused by the motion in a resonant

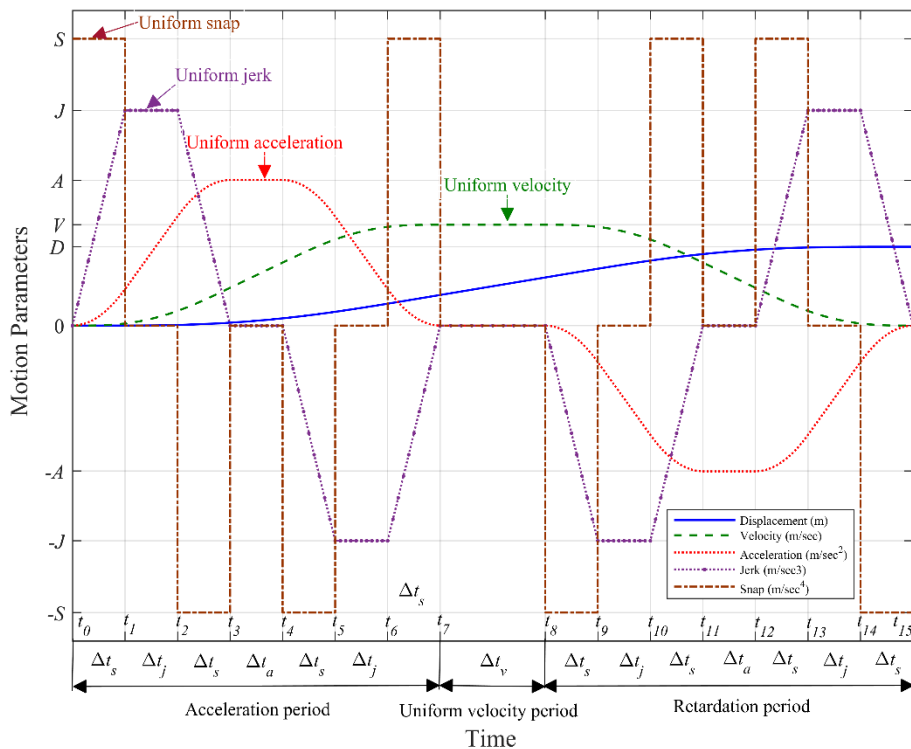


FIGURE 1. 15-SFOS S-curve profile showing the displacement, velocity, acceleration, jerk, and snap plotted with respect to time. The uniform time durations corresponding to each subplot are also indicated.

system [45], [46]. In [44], [45], and [46], optimization is carried out to develop the trajectories by considering both the kinematic constraints and the dynamic properties of the flexible link, thereby producing the minimum vibration response. The primary drawback of the trajectories described in [44], [45], and [46] is that they can suppress residual vibration, but this comes at the cost of a significantly increased TT. The TT is undefined and cannot be directly controlled, as it is determined according to the system's parameters. TT changes with any change in the system parameters. Furthermore, these optimization methods are highly detailed and require extensive system information, such as natural frequencies and damping coefficients, and are effective primarily when the system dynamics are well understood. A recent study addressed the minimization of residual vibration for a fixed TT without accounting for system dynamics [47], [48]. However, obtaining an optimal solution in this approach requires a large number of experiments, which is the major limitation of the study.

In most of the existing literature, the focus is on suppressing induced residual vibrations while simultaneously optimizing the TT. In contrast, the proposed approach aims to minimize residual vibrations for a specified TT. The following are the advantages of proposed method:

1. The CMP method is introduced to design a near-optimum 15-SFOS S-curve profile, aiming to achieve low-vibration P-T-P motion for a specific TT within the actuator capacities.

2. The CMP method enables the minimization of residual vibration response without the need to consider the dynamics of the system, thereby significantly reducing the effort involved in system analysis, which is one of the key advantages of this approach.
3. The presented method is simple and highly efficient since it allows for the optimization of the 15-SFOS S-curve profile using any of the three CMP: critical acceleration, critical jerk, and critical snap.
4. The 15-SFOS S-curve profile presented is of a higher order, providing increased smoothness, which allows for more efficient minimization of residual vibration compared to lower-order profiles.
5. The 15-SFOS S-curve profile is developed as a unitized profile, making it applicable for any specific TT and displacement.
6. The characteristics of the 15-SFOS S-curve profile are presented as nomograms, which are used to investigate the appropriate magnitude of MP, according to the preferred CMP, without requiring any calculations.

In this paper, the formulations for the 15-SFOS S-curve profile are initially presented. The characteristics of this profile are presented as nomograms created using a novel unitization method that treats TT and displacement as one unit each. According to the CMP, the near-optimum

magnitude of non-critical motion parameters (NCMP) can be easily determined without any calculations using the proposed nomograms. The selected unitized magnitude (UM) can be readily converted to actual magnitude (AM) using multiplication factors (MF) derived from actual displacement and actual TT. The efficacy of the proposed method is experimentally demonstrated on the setup of flexible links. Furthermore, the mathematical modeling of the flexible link is provided, serving as the basis for simulated results to validate the experimental findings. Finally, the feasibility and practicality of the presented optimum trajectory are evaluated through experimental studies, with a comparison made to other benchmark trajectories. The remainder of this paper is organized as follows: Section II outlines the methodology for creating a 15-SFOS S-curve profile, while Section III details the generation of this profile, considering kinematic limitations and vibration suppression capabilities. Section IV presents the modelling of flexible system. Section V details the experimental validation of the CMP method, and concluding remarks are provided in Section VI.

II. 15-SFOS S-CURVE TRAJECTORY PLANNING

The fourth-order S-curve profile proposed in this paper is a 15-SFOS S-curve profile. To ensure smoothness and continuity up to the fourth derivative of displacement, i.e., snap, a fourth-order S-curve requires additional constant-snap and constant-jerk intervals compared to a trapezoidal velocity motion profile. These intervals must be included in both the acceleration and deceleration phases. As a result, the profile exhibits a symmetrical structure with seven segments in the acceleration phase and seven segments in the deceleration phase, along with a single constant-velocity segment, giving a total of 15 segments. This 15-segment structure is therefore inherent to any fourth-order S-curve with continuous snap. Fig. 1 illustrates the schematic representation of the 15-SFOS profile, which includes displacement, velocity, acceleration, jerk, and snap trajectories. The time durations corresponding to segments of uniform snap, jerk, acceleration, and velocity are denoted by Δt_s , Δt_j , Δt_a , and Δt_v , respectively. This profile consists of eight Δt_s segments, four Δt_j segments, two Δt_a segments, and one Δt_v segment, giving the TT as $\Delta t = 8\Delta t_s + 4\Delta t_j + 2\Delta t_a + \Delta t_v$. In Fig. 1, the maximum magnitude of displacement is represented as D , while the peak magnitudes of velocity, acceleration, jerk, and snap are, respectively, denoted as V , A , J , and S , respectively.

Fig. 1. 15-SFOS S-curve profile showing the displacement, velocity, acceleration, jerk, and snap plotted with respect to time. The uniform time durations corresponding to each subplot are also indicated.

The generalized formulations of the 15-SFOS S-curve profile are established using general polynomial equations. The snap profile within the 15-SFOS S-curve is defined by the equation provided in Eq. (1). This profile is determined individually for each time segment, with J , Δt_s serving as

variables.

$$S(t) = \begin{cases} \frac{J}{\Delta t_s} & 0 \leq t \leq t_1, t_6 \leq t \leq t_7 \\ 0 & t_{10} \leq t \leq t_{11}, t_{12} \leq t \leq t_{13} \\ & t_1 \leq t \leq t_2, t_3 \leq t \leq t_4, t_5 \leq t \leq t_6, t_7 \leq t \leq t_8 \\ & t_9 \leq t \leq t_{10}, t_{11} \leq t \leq t_{12}, t_{13} \leq t \leq t_{14} \\ -\frac{J}{\Delta t_s} & t_2 \leq t \leq t_3, t_4 \leq t \leq t_5 \\ & t_8 \leq t \leq t_9, t_{14} \leq t \leq t_{15} \end{cases} \quad (1)$$

where $S(t)$ represents the snap at any given time.

Furthermore, the formulation for the jerk profile, $J(t)$, can be derived by performing a single integration of Eq. (1) for a specific time. The jerk profile is expressed in Eq. (2). Similarly, the acceleration profile $A(t)$, can be derived by integrating Eq. (2) once with respect to time. Additionally, the formulations for the velocity profile, $V(t)$, and the displacement profile, $D(t)$, can be derived by performing one and two integrations, respectively, of $A(t)$ for a given time. The resulting equations for A , V , and D derived from these integrations are expressed in Eqs. (3), (4), and (5), respectively, with J as the known parameter.

' t ' represents the time for which the given equations remain applicable.

$$J(t) = \begin{cases} \frac{Jt}{\Delta t_s} & t_0 \leq t \leq t_1 \\ J & t_1 \leq t \leq t_2, t_{13} \leq t \leq t_{14} \\ \frac{J(2\Delta t_s + \Delta t_j - t)}{\Delta t_s} & t_2 \leq t \leq t_3 \\ 0 & t_3 \leq t \leq t_4, t_7 \leq t \leq t_8, \\ & t_{11} \leq t \leq t_{12} \\ \frac{J(2\Delta t_s + \Delta t_j + \Delta t_a - t)}{\Delta t_s} & t_4 \leq t \leq t_5 \\ -J & t_5 \leq t \leq t_6, t_9 \leq t \leq t_{10} \\ \frac{J(t - 4\Delta t_s - 2\Delta t_j - \Delta t_a)}{\Delta t_s} & t_6 \leq t \leq t_7 \\ \frac{J(4\Delta t_s + 2\Delta t_j + \Delta t_a + \Delta t_v - t)}{\Delta t_s} & t_8 \leq t \leq t_9 \\ \frac{J(t - 6\Delta t_s - 3\Delta t_j - \Delta t_a - \Delta t_v)}{\Delta t_s} & t_{10} \leq t \leq t_{11} \\ \frac{J(t - 6\Delta t_s - 3\Delta t_j - 2\Delta t_a - \Delta t_v)}{\Delta t_s} & t_{12} \leq t \leq t_{13} \\ \frac{J(8\Delta t_s + 4\Delta t_j + 2\Delta t_a + \Delta t_v - t)}{\Delta t_s} & t_{14} \leq t \leq t_{15} \end{cases} \quad (2)$$

$$A = \begin{cases} J(\Delta t_s + \Delta t_j) & t \in [t_7, t_8] \\ J(-\Delta t_s - \Delta t_j) & t \in [t_{11}, t_{12}] \end{cases} \quad (3)$$

$$V = J(2\Delta t_s + \Delta t_j + \Delta t_a)(\Delta t_s + \Delta t_j) \quad t \in [t_7, t_8] \quad (4)$$

$$D = J(2\Delta t_s + \Delta t_j + \Delta t_a)(\Delta t_s + \Delta t_j)$$

$$(4\Delta t_s + 2\Delta t_j + \Delta t_a + \Delta t_v) \quad t \in t_8 \quad (5)$$

By substituting Eq. (5) into Eqs. (1) and (2), the formulations for $S(t)$ and $J(t)$ with D , Δt_v , Δt_a , Δt_j , and Δt_s as known parameters are established. The formulas for Δt_v , Δt_a , Δt_j , and Δt_s as a function of D , A , V , J , and S are as follows:

$$\Delta t_s = \frac{J}{S} \quad (6)$$

$$\Delta t_j = \frac{AS - J^2}{SJ} \quad (7)$$

$$\Delta t_a = \frac{VJS - A^2S - AJ^2}{AJS} \quad (8)$$

$$\Delta t_v = \frac{DAJS - A^2VS - VAJ^2 - V^2JS}{AVJS} \quad (9)$$

A. UNITIZATION OF 15-SFOS S-CURVE PROFILE

A novel unitization method is proposed to generalize the profile by considering displacement and TT as one unit ($D = \Delta t = 1$). The resulting 15-SFOS S-curve profile generated by this method is referred to as the US-curve profile, with the magnitudes of its MP denoted as UM. These UM values can be converted to actual dimensions by multiplying them by the motion factors (MF) listed in Table 1, where K_1 represents the actual displacement, and K_2 represents the actual TT. The MF_v , MF_a , MF_j , and MF_s correspond to the MF for velocity, acceleration, jerk, and snap, respectively.

The conventional uniform velocity, uniform acceleration, uniform jerk, and uniform snap profiles can be seen as special cases of the 15-SFOS S-curve profile. These profiles achieve the minimum possible peak magnitudes for the MP. Table 2 provides the specific values of these minimum peak magnitudes, along with the corresponding Δt_s , Δt_j , Δt_a , and Δt_v for the US-curve profile, for easy reference. These values are also illustrated in Fig. 2. Just as the minimum velocity magnitude is attained with the uniform velocity profile, as shown in Fig. 2(a), the minimum peak magnitudes of acceleration, jerk, and snap can similarly be achieved using the respective uniform profiles, as depicted in Fig. 2(b–d). The magnitudes of the other components of the MP when achieving these minimum possible values are also detailed in Table 2.

TABLE 1. MF to obtain AM.

MP	Velocity (MF_v)	Acceleration (MF_a)	Jerk (MF_j)	Snap (MF_s)
MF	$\frac{K_1}{K_2}$	$\frac{K_1}{K_2^2}$	$\frac{K_1}{K_2^3}$	$\frac{K_1}{K_2^4}$

B. CHARACTERISTICS OF US-CURVE PROFILE

The contribution of this unitization method lies in its characteristics, which provide a general solution that can be easily used to achieve low-vibration P-T-P motion. The desired solution can be selected directly from the plots generated by this method. As shown in Figs. 3 and 4, these plots display

TABLE 2. Minimum possible magnitudes and corresponding magnitudes of other MP along with their time durations for US-curve profile.

	MP	V	A	J	S
	Magnitude	1	4	32	512
Corresponding to other peak magnitudes for minimum possible magnitude condition of the MP	V	1	2	2	2
	A	∞	4	8	8
	J	∞	∞	32	64
	S	∞	∞	∞	512
Corresponding time durations of segments	Δt_s	0	0	0	0.125
	Δt_j	0	0	0.25	0
	Δt_a	0	0.5	0	0
	Δt_v	1	0	0	0

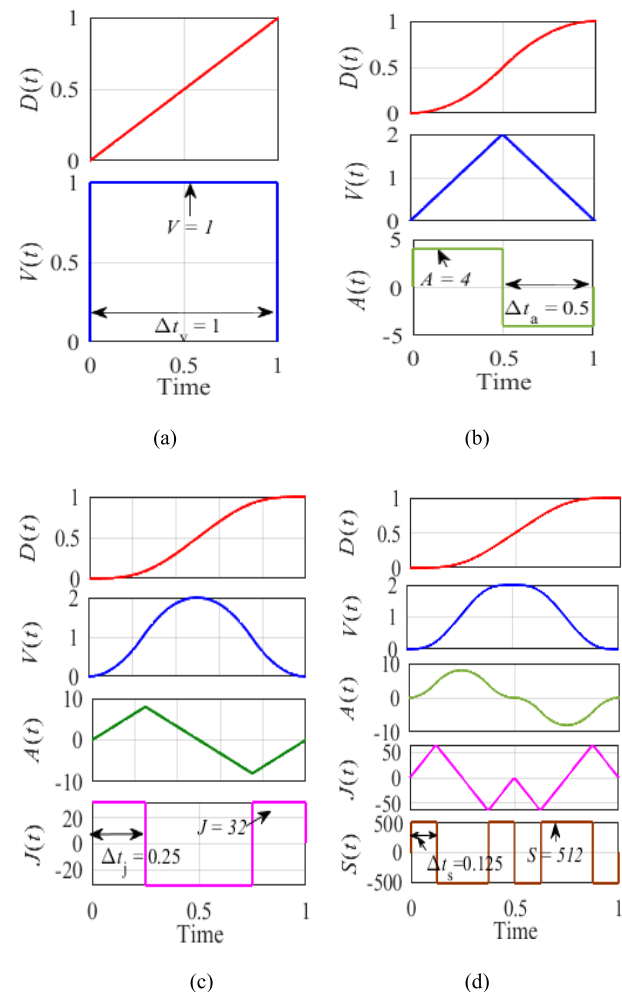


FIGURE 2. Conventional motion profiles: (a) uniform velocity with displacement and velocity plots; (b) uniform acceleration with displacement, velocity, and acceleration plots; (c) uniform jerk with displacement, velocity, acceleration, and jerk plots; and (d) uniform snap with displacement, velocity, acceleration, jerk, and snap plots.

all possible combinations of MP magnitudes that can be used to design a feasible 15-SFOS S-curve profile. Not only does the 15-SFOS curve profile, but also these graphs, allow observation of the possible combinations for designing a feasible

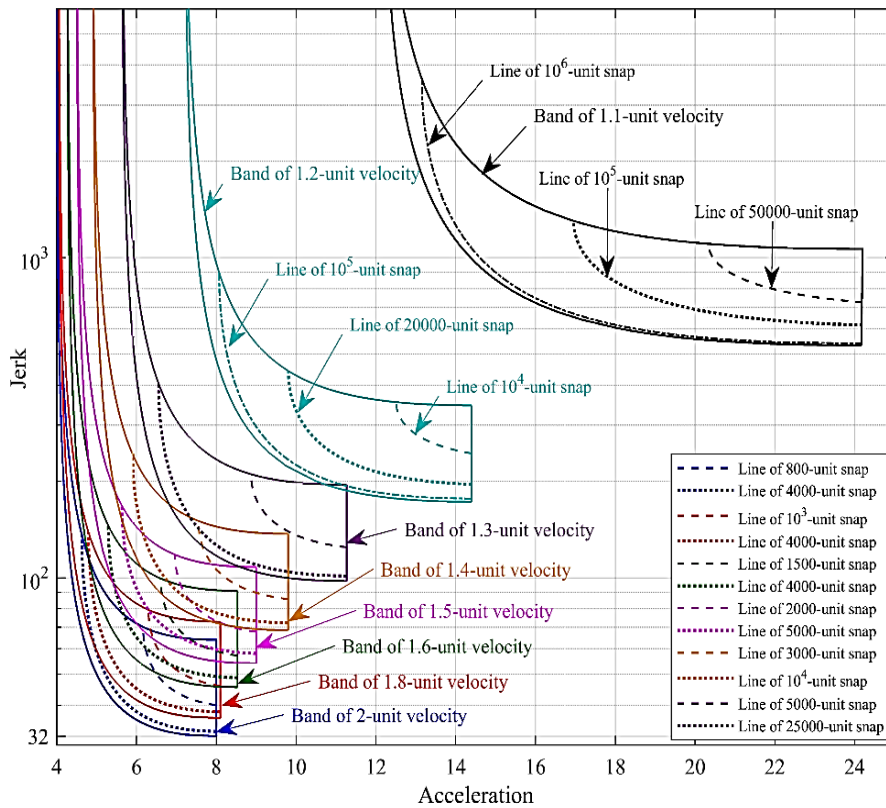


FIGURE 3. Unitized magnitude (UM) relationship between peak acceleration and peak jerk for different peak velocities, shown as bands. Within each peak-velocity band, the internal curves represent different levels of peak snap.

third-order S-curve profile. Using these graphs, the feasible combinations of UM of MP can be directly selected without any calculations by considering the given limiting magnitudes of MP, such as maximum velocity (V_{max}), maximum acceleration (A_{max}), maximum jerk (J_{max}), and maximum snap (S_{max}). The selected UM from these graphs can easily be converted to actual dimensions by multiplying the UM by the multiplication factors given in Table 1.

The graph shown in Fig. 3 is plotted between A and J , in which various bands of potential combinations of different V ranging from 1.1 to 2 units are depicted, with lines representing corresponding S magnitudes. Since some of the bands in Fig. 3 overlap, Fig. 4 displays these bands separately with more S lines to provide a clearer representation of the magnitudes of all the MP. The bottom curve of each band represents the magnitudes of A and J for the third-order S-curve profile, as it corresponds to the case of an infinite S . Fig. 4 shows the graphs for V between 1.3 and 2 units, while plots for V below 1.3 units are not included. This is because, when V drops below 1.3 units, the magnitudes of the other MP increase sharply, which is not desirable for practical implementations.

In the US-curve profile, A , J , and S can increase to infinity, while V can only reach twice of its minimum possible magnitude, as specified in Table 2. It's important to note

that as V increases, the magnitudes of the other MP tend to remain relatively lower. Therefore, when V is limited to 2 units, the magnitudes of the remaining motion parameters naturally decrease. However, the limiting magnitudes of MP, TT, and displacement all play a crucial role in maintaining a V of 2 units. If the limiting magnitude of MP do not allow maintaining V of 2 units, a lower magnitude of V can be selected using the provided nomograms (Fig. 3 and Fig. 4). This approach aligns with other concepts and serves as a selection criterion for MP magnitudes aimed at minimizing residual vibration, as explained in the following section.

C. NUMERICAL EXAMPLE

A numerical example is presented below to demonstrate the utility of the unitization concept. Table 3 lists the minimum possible peak UM of different MP in the second row, while the corresponding MF are given in the third row. The K_1 and K_2 are taken as 1.2 metre and 2 sec, respectively. The fourth row lists the minimum possible peak AM of the MP, which are obtained by multiplying the minimum possible peak UM by the corresponding MF.

Let us consider the actuator constraints $V_{max} = 0.96$ m/s (UM = 1.6 units) and $A_{max} = 1.92$ m/s² (UM = 6.4 units). If a minimum magnitude of S is required, the approximate UM of S and J can be selected from Fig. 5 as 1640 units

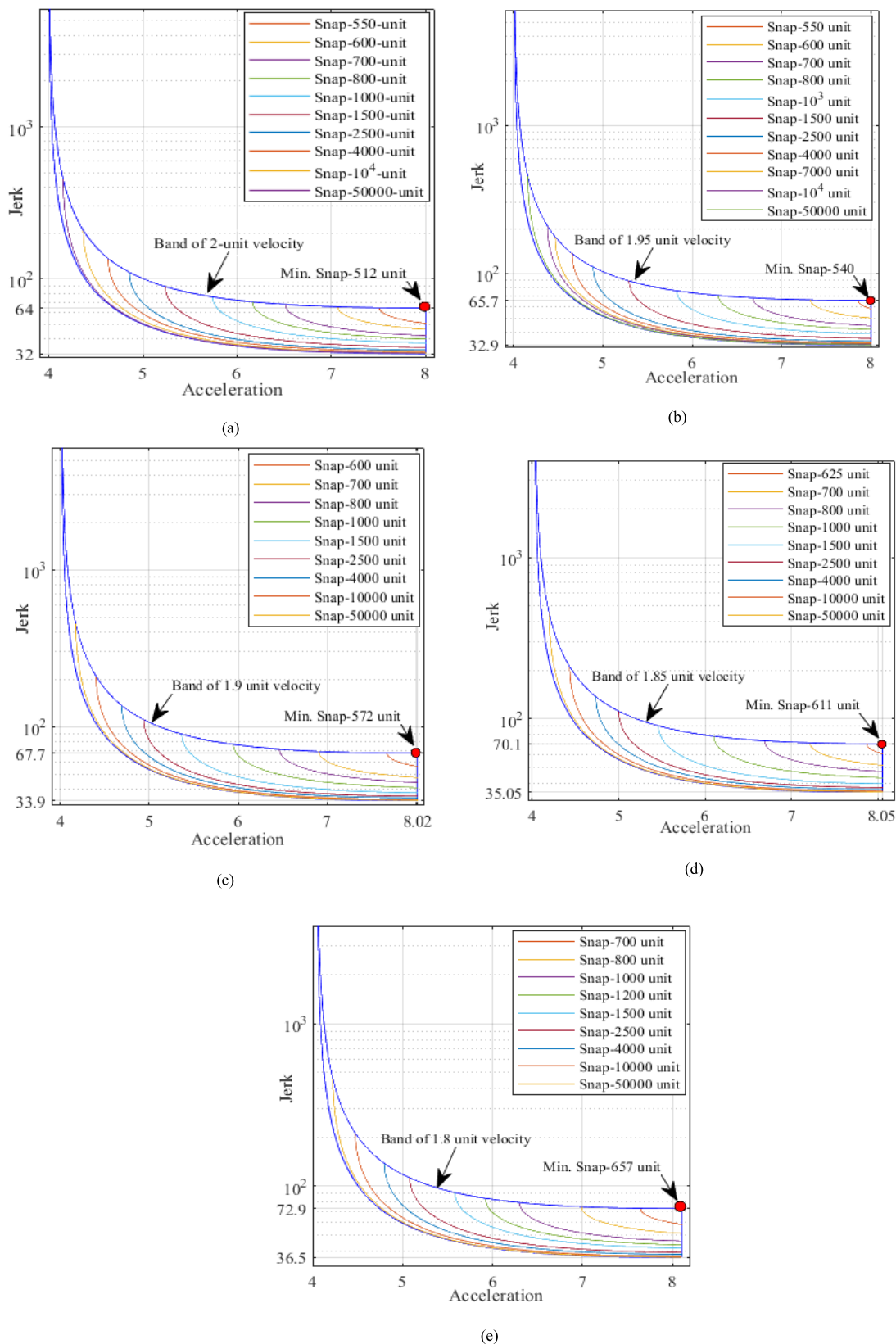


FIGURE 4. Unitized magnitude (UM) relationships between peak acceleration and peak jerk for different unitized peak velocities, shown as bands in each subplot: (a) 2.0, (b) 1.95, (c) 1.90, (d) 1.85, (e) 1.80, (f) 1.75, (g) 1.70, (h) 1.65, (i) 1.60, (j) 1.50, (k) 1.40, and (l) 1.30 units. Within each peak-velocity band, the internal curves represent different levels of peak unitized snap.

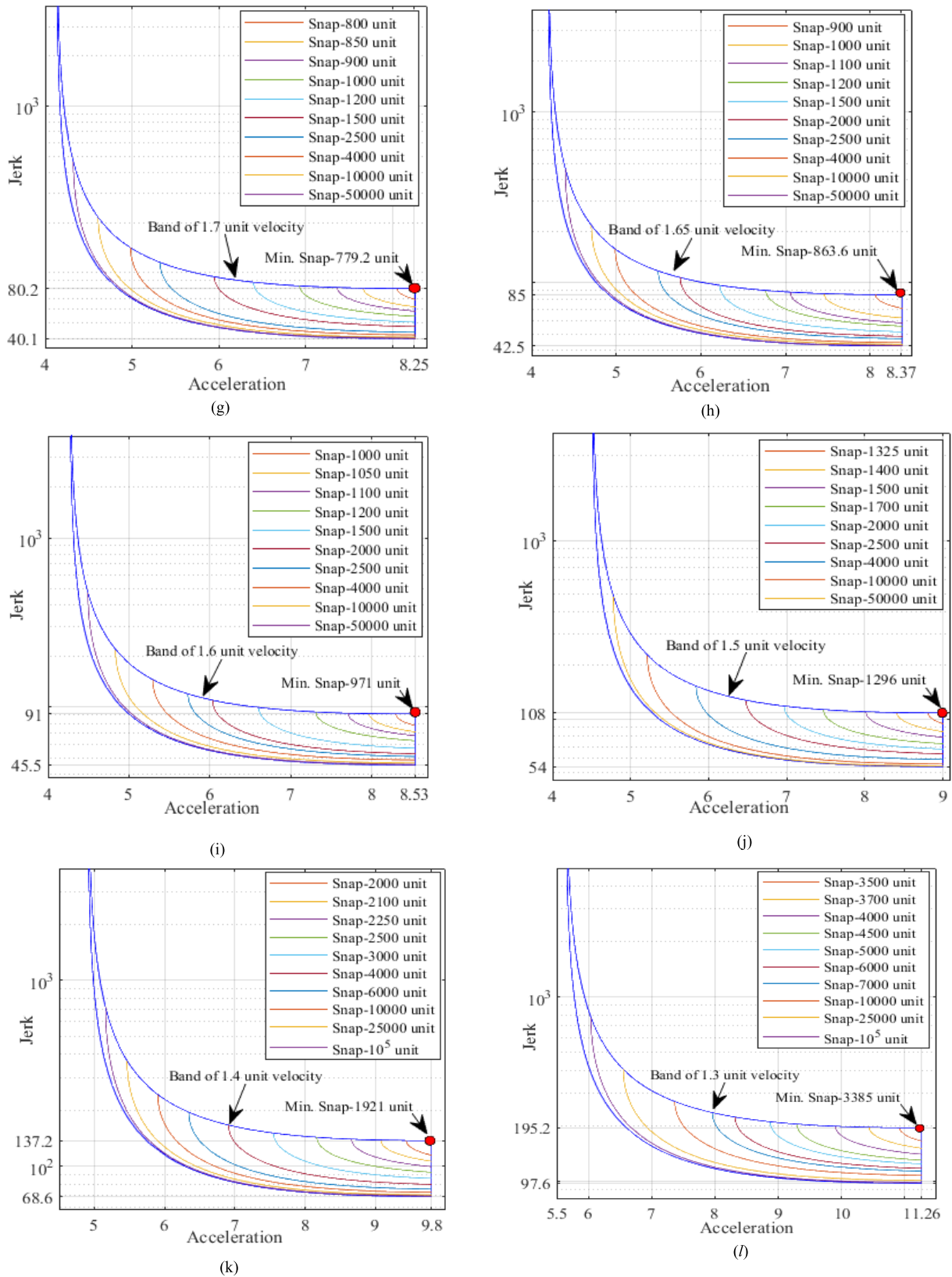


FIGURE 4. (Continued.) Unitized magnitude (UM) relationships between peak acceleration and peak jerk for different unitized peak velocities, shown as bands in each subplot: (a) 2.0, (b) 1.95, (c) 1.90, (d) 1.85, (e) 1.80, (f) 1.75, (g) 1.70, (h) 1.65, (i) 1.60, (j) 1.50, (k) 1.40, and (l) 1.30 units. Within each peak-velocity band, the internal curves represent different levels of peak unitized snap.

TABLE 3. Magnitudes of MP for a given numerical example.

MP	V	A	J	S
Minimum peak UM	1	4	32	512
MF	$\frac{K_1}{K_2}$ (0.6)	$\frac{K_1}{K_2^2}$ (0.3)	$\frac{K_1}{K_2^3}$ (0.15)	$\frac{K_1}{K_2^4}$ (0.075)
Minimum peak AM	0.6 m/sec	1.2 m/sec ²	4.8 m/sec ³	38.4 m/sec ⁴

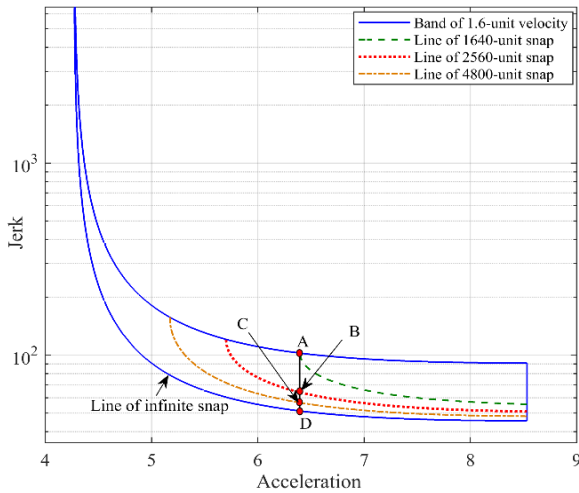


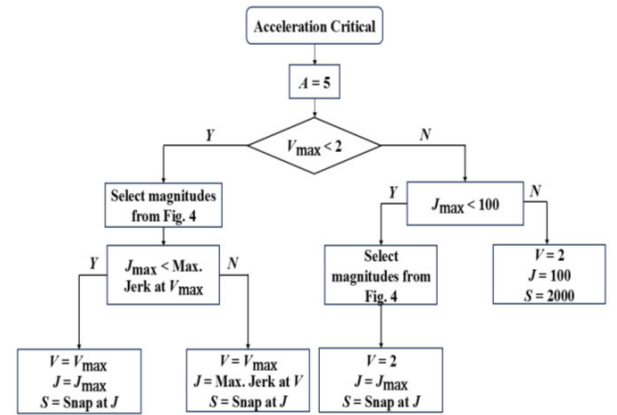
FIGURE 5. Unitized magnitude relationship between peak acceleration and peak jerk for a unitized peak velocity of 1.6 units, shown as a band. Within this band, the internal curves represent different levels of peak unitized snap.

and 102 units, respectively. Point A in Fig. 5 represents these magnitudes. The corresponding AM of S and J are calculated as 123 m/s⁴ and 15.3 m/s³, respectively. If the magnitude of S is ignored and only the minimum magnitude of J is required, the fourth-order trajectory switches to a third-order trajectory. This occurs because, at the minimum magnitude of J , the magnitude of S tends to infinity. The minimum UM of J can be visualized at point D in Fig. 5 as 51 units, and the corresponding AM of J is calculated as 7.65 m/s³.

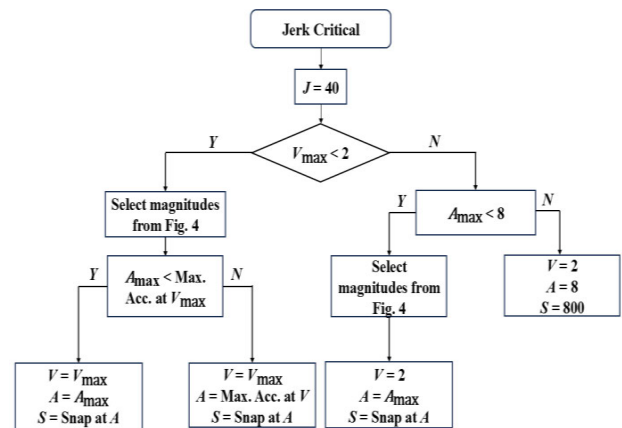
Now, let's also consider the third actuator limitation $J_{max} = 9.6$ m/s³ (UM = 64), along with V_{max} and A_{max} . For these limiting magnitudes of V , A , and J , the minimum UM of S is obtained from Fig. 5 as 2560 units, which gives an AM of 192 m/s⁴. These magnitudes are represented by point B in Fig. 5. Alternatively, if the limiting magnitude of S is specified as 360 m/s⁴ (UM = 4800) instead of a limiting magnitude of J , and the minimum magnitude of J is required, Fig. 5 provides the UM of J as 56 units at point C, resulting in an AM of 8.4 m/s³.

III. GENERATION OF S-CURVE PROFILE WITH KINEMATIC LIMITATIONS AND VIBRATION SUPPRESSION CAPABILITIES

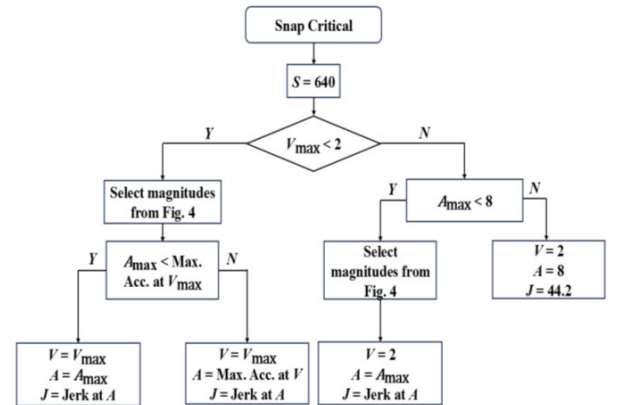
The control of residual vibration amplitude depends not only on the smoothness of the profile but also on the selection of the appropriate magnitude of MP. Smoothness can be



(a)



(b)



(c)

FIGURE 6. Flowchart for selecting optimum magnitudes of motion parameters when (a) acceleration is critical, (b) jerk is critical and (c) snap is critical.

achieved and further enhanced by increasing the order of the profile. The 15-SFOS S-curve profile provides high smoothness for controlling residual vibration.

In most literature, the vibration problem is solved using the time-optimal profile, where TT is determined based on the given limiting magnitudes of MP. In this time-optimal profile,

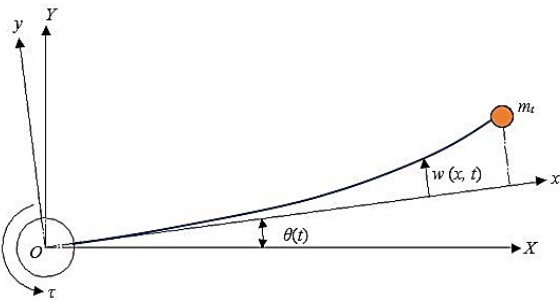


FIGURE 7. Modelling of flexible rotating link.

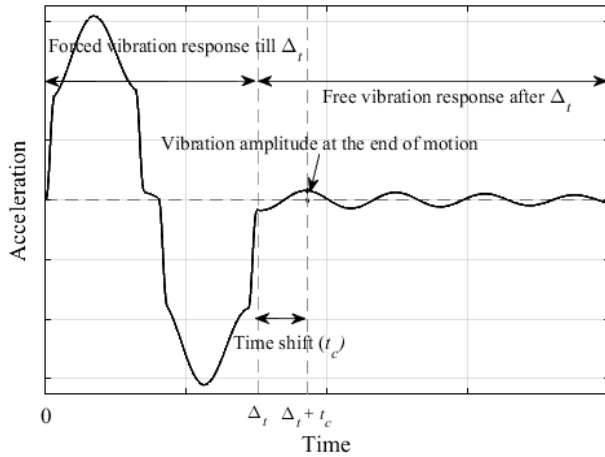


FIGURE 8. Theoretical response of residual vibration in terms of acceleration at the tip end of the flexible rotating link.

the magnitudes of the MP, when feasible, are set equal to their limiting values. If the given limiting magnitudes are not feasible, an algorithm is used to verify the feasibility conditions by progressively reducing the magnitudes of MP below their limiting magnitudes. This time-optimal profile can execute the P-T-P motion in the minimum TT; however, it does not guarantee effective suppression of residual vibration when the system dynamics vary. This is because the profile is designed according to the limiting magnitudes of MP and does not consider the importance of each MP relative to the system parameters. It is crucial to consider the importance of each MP in designing the profile to achieve low-vibration P-T-P motion. Therefore, the CMP method is introduced to determine the near-optimum magnitude of MP for achieving low-vibration P-T-P motion for a specific TT and distance traveled without addressing the system dynamics. This consideration leads to a general solution that can be applied to any system without the need for its dynamic equations and dynamic parameters.

To achieve low-vibration P-T-P motion using the presented CMP method, near-optimum profiles are designed by treating acceleration, jerk, and snap individually as CMP. This method generates a motion profile in a manner similar to the time-optimal profile. The key distinction is that the magnitude of the CMP is selected to be close to its minimum feasible value, while the magnitudes of the other MP or NCMP, when

TABLE 4. Parameters of the experimental setup.

Parameters	Values
ω_1	3.49 Hz
ω_2	5.64 Hz
ζ_1	0.01
ζ_2	0.01
m_1	0.123 kg/m
m_2	0.177 kg/m
D	0.62 m (2.094 rad)
$V_{\max}$	1.776 m/sec (6 rad/sec)
$A_{\max}$	14.8 m/sec ² (50 rad/sec ²)
$J_{\max}$	296 m/sec ³ (1000 rad/sec ³)
$S_{\max}$	7400 m/sec ⁴ (25000 rad/sec ⁴)

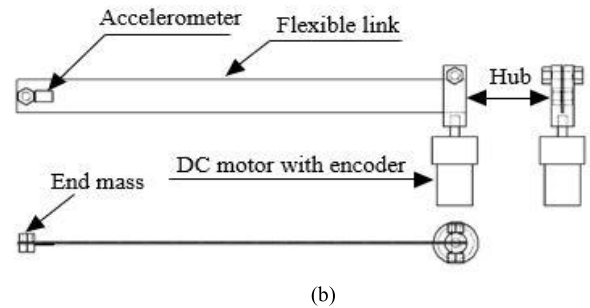
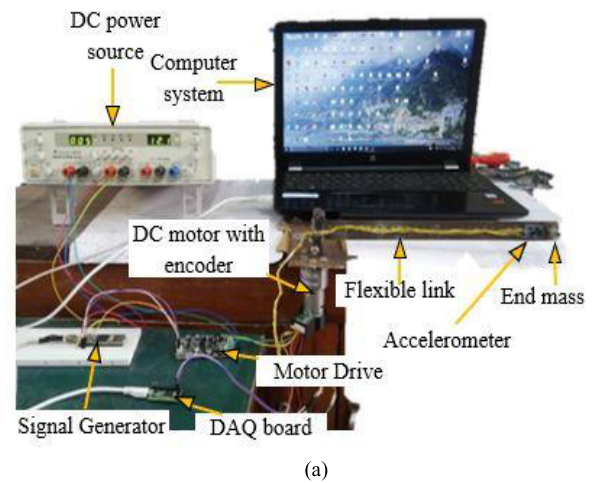


FIGURE 9. (a) Experimental setup (b) schematic of flexible rotating link.

feasible, are set equal to their respective limiting values. It is also beneficial to maintain the magnitude of CMP equal to its minimum value. This approach results in the creation of conventional uniform profiles, as depicted in Fig. 2. These uniform profiles are commonly used in electromechanical P-T-P motion applications. However, like the time-optimal profile, these uniform profiles are not highly accurate in suppressing the residual vibration. Additionally, because these are uniform profiles, it is not possible to maintain fourth-order continuity in all of them; only the uniform snap profile is bounded with an order of four. Therefore, the magnitude of CMP is slightly increased by around 25% over the

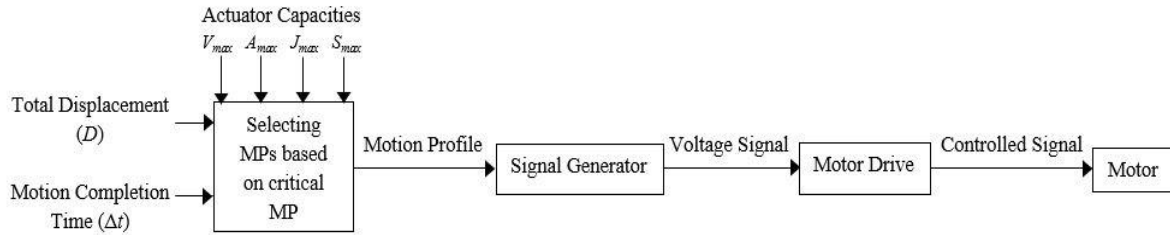


FIGURE 10. Simplified control block diagram.

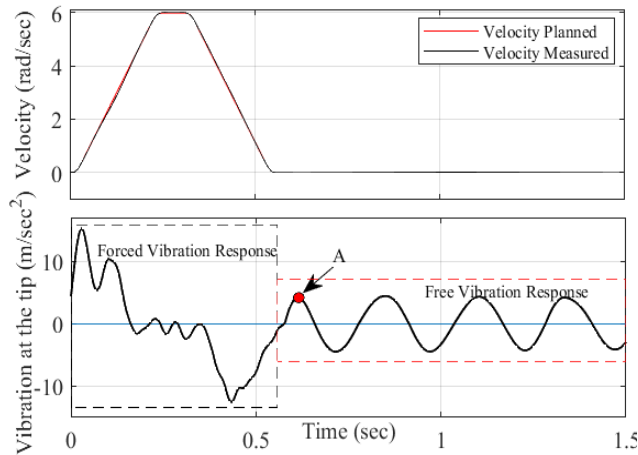


FIGURE 11. Response at the tip of the 1st link for a planned velocity profile with a minimum TT of 0.559 seconds.

minimum possible magnitude, which bounds the profiles with order four while also keeping the magnitude of CMP close to the minimum possible value. This consideration leads to near-optimum profiles that possess the characteristics of both the time-optimal profile and conventional uniform profiles. With this increment, the A is raised from 4 to 5 units, the J is increased from 32 to 40 units, and the S is increased from 512 to 640 units. In practice, continuously tuning the profile with a required CMP and other lower magnitudes of CMP (other than 5, 40, and 640 units) to accurately suppress the residual vibration modes as the system dynamics change is very difficult. Therefore, to achieve low-vibration P-T-P motion, profiles with acceleration, jerk, and snap as the CMP, each set 25% above their minimum possible values, need to be used. While this may not provide the accurate optimum solution, it can provide a near-optimum solution for achieving low-vibration P-T-P motion.

For the given values of CMP magnitudes, the magnitudes of NCMP can be determined using the nomograms depicted in Fig. 4. When considering acceleration as a CMP with $A = 5$ units, the magnitude of V is equal to its maximum possible value, i.e., $V = 2$ unit, with the condition that $2 \leq V_{max}$. If $2 \leq V_{max}$, the magnitude of J is set to 100 units, which is the maximum possible value of J for $A = 5$ and $V = 2$ units, as indicated in the nomogram presented in Fig. 4(a).

TABLE 5. MF for different TT.

TT \ MF	0.75 sec	1 sec	1.25 sec	1.5 sec	1.75 sec	2 sec
MF _v	2.79	2.094	1.675	1.396	1.196	1.047
MF _a	3.72	2.094	1.34	0.931	0.683	0.523
MF _j	4.96	2.094	1.072	0.62	0.391	0.262
MF _s	6.62	2.094	0.857	0.413	0.223	0.131

TABLE 6. ULM of MP for different TT.

TT \ ULM	0.75 sec	1 sec	1.25 sec	1.5 sec	1.75 sec	2 sec
UL M _v	2.14	2.86	3.58	4.29	5.01	5.73
UL M _a	13.4	23.87	37.31	53.7	73.2	95.6
UL M _j	201	477	933	1613	2557	3817
UL M _s	3777	11939	29171	60533	112108	190840

When $A = 5$, $V = 2$, and $J = 100$ units, the magnitude of S is determined to be 2000 units. If the magnitude of $J = 100$ units exceed J_{max} , then J is set to J_{max} ($J = J_{max}$). Under this condition, the magnitude of S can be easily obtained from the nomogram provided in Fig. 4(a). Furthermore, if $V = 2$ exceeds V_{max} , the magnitude of V is adjusted to its next lower value very close to V_{max} , as per the nomograms shown in Fig. 4(b)-4(l). In this scenario, the magnitude of J is its maximum possible value, which can be determined from Fig. 4(b)-4(l), with the condition that $J \leq J_{max}$. Subsequently, the magnitude of S can also be readily obtained from the same graphs. If $J > J_{max}$, then J is set to J_{max} ($J = J_{max}$), and the magnitude of S can be conveniently determined from the graphs presented in Fig. 4(b)-4(l).

When considering jerk as a CMP with $J = 40$ units, the magnitude of V is equal to its maximum possible magnitude, i.e., $V = 2$ units, with a condition that $2 \leq V_{max}$. If $2 \leq V_{max}$, the magnitude of A is set to 8 units, which is the maximum possible value of A for $J = 40$ and $V = 2$ units, as indicated in the nomogram presented in Fig. 4(a). When $J = 40$,

TABLE 7. Optimum UM and AM of MP based on CMP for different TT.

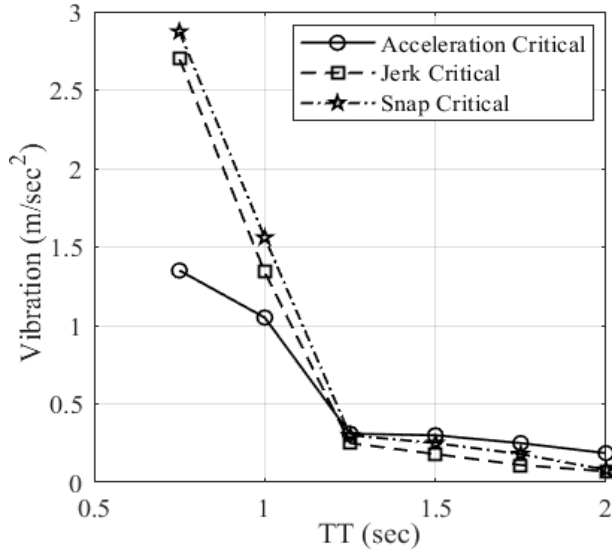
CMP	TT (sec) MP		0.75	1	1.25	1.5	1.75	2
Acc.	V	UM (units)	2	2	2	2	2	2
		AM (rad/sec)	5.58	4.18	3.35	2.79	2.39	2.09
	A	UM (units)	5	5	5	5	5	5
		AM (rad/sec ²)	18.6	10.4 7	6.7	4.65	3.42	2.62
	J	UM (units)	100	100	100	100	100	100
		AM (rad/sec ³)	496	209. 4	107	62	39.07	26.2
	S	UM (units)	2000	2000	2000	2000	2000	2000
		AM (rad/sec ⁴)	13236	4188	1715	827	446.5	261. 7
Jerk	V	UM (units)	2	2	2	2	2	2
		AM (rad/sec)	5.58	4.18	3.35	2.79	2.39	2.09 4
	A	UM (units)	8	8	8	8	8	8
		AM (rad/sec ²)	29.78	16.7 5	10.7	7.44	5.47	4.18 8
	J	UM (units)	40	40	40	40	40	40
		AM (rad/sec ³)	198.54	83.7 6	42.8	24.8	15.63	10.4 7
	S	UM (units)	800	800	800	800	800	800
		AM (rad/sec ⁴)	5294.4	1675	686	330. 9	178.6	104. 7
Snap	V	UM (units)	2	2	2	2	2	2
		AM (rad/sec)	5.58	4.19	3.35	2.79	2.39	2.09
	A	UM (units)	8	8	8	8	8	8
		AM (rad/sec ²)	29.78	16.7 5	10.7	7.44	5.47	4.18
	J	UM (units)	44.2	44.2	44.2	44.2	44.2	44.2
		AM (rad/sec ³)	219.5	92.6	47.4	27.4	17.28	11.5 7
	S	UM (units)	640	640	640	640	640	640
		AM (rad/sec ⁴)	4235.5	1340	549	264. 7	142.89	83.7 6

$V = 2$ and $A = 8$ units, the magnitude of S is determined to be 800 units. If the magnitude of $A = 8$ units exceeds A_{max} , the magnitude of A is set to A_{max} ($A = A_{max}$), and for this condition the magnitude of S can be easily obtained from the nomogram provided in Fig. 4(a). Furthermore, if $V = 2$ exceed V_{max} , the magnitude of V is adjusted to its next lower value very close to V_{max} , as per the nomograms shown in Fig. 4(b-l). For this condition, the magnitude of A is considered equal to its maximum possible value, which can be determined from Fig. 4(b-l), with the condition that $A \leq A_{max}$. Subsequently, the magnitude of S can also be obtained from the same graphs. If $A > A_{max}$, then A is set to A_{max} ($A = A_{max}$), and the magnitude of S can be conveniently determined from the graphs presented in Fig. 4(b-l).

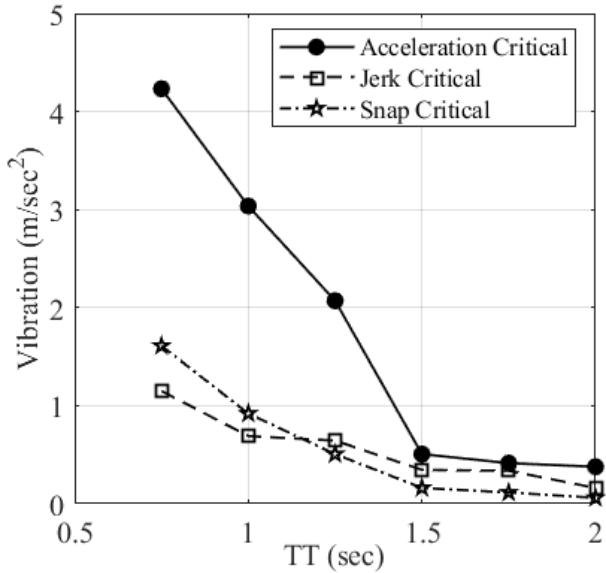
When considering snap as a CMP with $S = 640$ units, the magnitude of V is considered equal to its maximum possible magnitude, i.e., $V = 2$ units, with a condition that $2 \leq V_{max}$. If $2 \leq V_{max}$, the magnitude of A is set to 8 units, which is the maximum possible value of A for $S = 640$ and

$V = 2$ units, as indicated in the nomogram presented in Fig. 4(a). When $S = 640$, $V = 2$ and $A = 8$ units, the magnitude of J is determined to be 4.2 units. If the magnitude of $A = 8$ units exceeds A_{max} , the magnitude of A is set to A_{max} ($A = A_{max}$). For this condition, the magnitude of J can be easily obtained from the nomogram provided in Fig. 4(a). Furthermore, if $V = 2$ exceeds V_{max} , the magnitude of V is adjusted to its next lower value very close to V_{max} , as per the nomograms shown in Fig. 4(b-l). For this condition, the magnitude of A is considered to be equal to its maximum possible value, which can be obtained from Fig. 4(b-l), with a condition that $A \leq A_{max}$. Subsequently, the magnitude of J can also be determined from the same graphs. If $A > A_{max}$, the magnitude of A is set to A_{max} ($A = A_{max}$), and the magnitude of J can be easily determined from the graphs shown in Fig. 4(b-l).

If the magnitude of $V = 2$ units exceeds V_{max} , then V is selected based on the following conditions. Depending on this condition, a specific graph is chosen from which the



(a)



(b)

FIGURE 12. Residual vibration amplitude with respect to time at the end of P-T-P motion when acceleration, jerk, and snap are made critical for (a) the first link and (b) the second link.

magnitudes of other MP can be easily determined without any additional calculations. For the condition where $2 > V_{max} \geq 1.95$, the graph shown in Fig. 4(b) is chosen for further findings. Similarly, for the conditions $1.95 > V_{max} \geq 1.9$, $1.9 > V_{max} \geq 1.85$, $1.85 > V_{max} \geq 1.8$, $1.8 > V_{max} \geq 1.75$, $1.75 > V_{max} \geq 1.7$, $1.7 > V_{max} \geq 1.65$, $1.65 > V_{max} \geq 1.6$, $1.6 > V_{max} \geq 1.5$, $1.5 > V_{max} \geq 1.4$, and $1.4 > V_{max} \geq 1.3$, the graphs shown in Figs. 4(c), 4(d), 4(e), 4(f), 4(g), 4(h), 4(i), 4(j), 4(k), and 4(l), are respectively selected for further findings.

All conditions for determining the near-optimum magnitudes for a specific CMP are presented clearly in the form of flowcharts in Fig. 6.

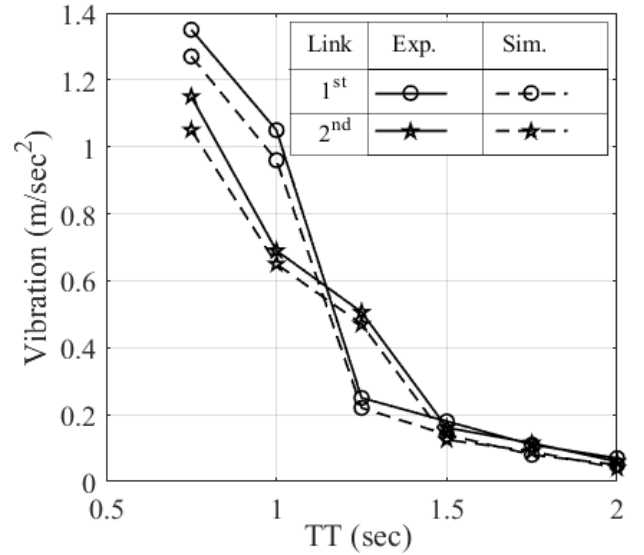


FIGURE 13. Comparison of simulated and experimental minimum residual vibration amplitudes with respect to time at the end of P-T-P motion.

An underdamped flexible rotating link is used for P-T-P motion to demonstrate the effectiveness of the presented CMP method on a flexible system. Such a system can exhibit residual vibrations both during and after the motion. The primary objective is to minimize the residual vibration at the end of the P-T-P motion. The theoretical background and experimental details of this system are presented in the following sections.

IV. MODELING OF FLEXIBLE SYSTEM

The flexible rotating link considered in this study is modeled a Euler-Bernoulli beam. The schematic representation is shown in Fig. 7, where $\theta(t)$ denotes the angular displacement of the hub, m_t represents the mass attached at the free end, and $w(x, t)$ indicates the transverse deflection of the link at position x and time t . A dynamic model formulation for extracting the residual vibration response is developed using Lagrange's equation. The deflection of the flexible link, $w(x, t)$, is expressed using an assumed mode method as follows:

$$w(x, t) = \sum_{i=1}^{\infty} \phi_i(x) \eta_i(t) \quad (10)$$

where, $\phi_i(x)$ and $\eta_i(t)$ represent the normalized mode shape function and the time-dependent modal coordinate corresponding to the i th harmonic mode, respectively. Since the first natural frequency of the flexible link is dominant, only the first mode is considered to approximate the elastic deflection [49].

$$w(x, t) = \phi(x) \eta(t) \quad (11)$$

where, $\eta(t) = \eta_1(t)$ represents the first-time dependent elastic mode and $\phi(x) = \phi_1(x)$ represents the first normalized mode for the fixed-free boundary condition.

TABLE 8. Performance comparison of the various trajectory model on vibration.

Work	Trajectory Model	Method Used	Initial consideration	Link	TT (sec)	Amplitude of vibration (m/sec ²)	System dynamics and Parameters
Lambrechts et al. [42], Fan et al. [43]	Traditional Fourth-order S-curve	Time-optimal trajectory planning method	Moving distance (D), Actuator capacities ($V_{\max}, A_{\max}, J_{\max}, S_{\max}$)	1 st	0.559	4.22	Not required
				2 nd		6.51	
Lee and Ha [44]	Traditional Optimum Fourth-order S-curve	General optimization method based on Laplace domain approach	Moving distance (D), Actuator capacities ($V_{\max}, A_{\max}$), Natural frequency (ω), Damping ratio (ζ)	1 st	2.292	0.052	Required (Natural frequency, Damping ratio)
				2 nd	1.418	0.142	
Proposed	Optimum 15-SFOS S-curve	Optimization using CMP method	Moving distance (D), TT (Δt), Actuator capacities ($V_{\max}, A_{\max}, J_{\max}, S_{\max}$)	1 st	0.75	1.35	Not required
				2 nd		1.15	
				1 st	1	1.05	
				2 nd		0.69	
				1 st	1.25	0.25	
				2 nd		0.506	
				1 st	1.5	0.18	
				2 nd		0.161	
				1 st	1.75	0.11	
				2 nd		0.115	
				1 st	2	0.071	
				2 nd		0.062	

The simplified dynamic equation of the flexible rotating link, which is derived using Lagrange's equation, is presented as follows [50]:

$$m_1 \ddot{\theta}(t) + m_2 \ddot{\eta}(t) + c \dot{\eta}(t) + k \eta(t) = 0 \quad (12)$$

where, c represents the damping coefficient ($c = 2\xi\sqrt{mk}$, ξ is the damping ratio), k represents the stiffness factor and is calculated as $k = m\omega^2$, where m stands for the mass per unit length (kg/m), and ω is the dominant natural frequency in Hz of the flexible link.

$$m_1 = m \int_0^l \phi(x) x dx + m_l l \phi(l) \quad (13)$$

$$m_2 = m + m_l \phi^2(l) \quad (14)$$

The resulting simplified transfer function of the Eq. (12) for the deflection $w(l, t)$, and relative acceleration $\ddot{w}(l, t)$ at the tip end ($x = l$) in the Laplace domain is as follows:

$$\frac{w(s)}{\theta(s)} = \frac{\ddot{w}(s)}{\ddot{\theta}(s)} = \frac{-m_1 \phi(l) s^2}{m_2 s^2 + 2sm\omega\xi + m\omega^2} \quad (15)$$

where, $\theta(s)$, $\ddot{\theta}(s)$, $w(s)$ and $\ddot{w}(s)$ are the Laplace forms of $D(t)$, $A(t)$, $w(l, t)$ and $\ddot{w}(l, t)$, respectively.

To align with the profile, the relationship is established as $\theta(t) = D(t)$, and their Laplace transforms are denoted as $\theta(s) = D(s)$. The Laplace transform of the displacement profile, $D(s)$, for the 15-SFOS S-curve profile, can be expressed as follows:

$$D(s) = \frac{1}{s^5} \sum_{i=0}^{15} S_i e^{-t_i s} \quad (16)$$

S_i represents the magnitude of the peak Snap at t_i . The tip deflection of the link is obtained by applying the inverse Laplace transform after substituting Eq. (16) into Eq. (15).

$$w(t) = -m_1 \phi(l) \sum_{i=0}^{15} S_i Z_i H_i \quad (17)$$

Similarly, the relative acceleration of the link at the tip end is given by the following equation:

$$\ddot{w}(t) = -m_1 \phi(l) \sum_{i=0}^{15} S_i Y_i H_i \quad (18)$$

where,

$$Z_i = \frac{(t - t_i)^2}{2k} + \frac{A_i C_i (3m_2 - c^2) c}{Dk^3} + \frac{B_i C_i (km_2 - c^2) + c^2}{k^3} - \frac{c(t - t_i) + m_2}{k^2}$$

$$Y_i = \frac{1}{k} - \frac{A_i C_i c}{Dk} \frac{B_i C_i}{k}$$

$$A_i = \sinh\left(\frac{(t - t_i) D}{2m_2}\right), \quad B_i = \cosh\left(\frac{(t - t_i) D}{2m_2}\right),$$

$$C_i = e^{-\frac{(t - t_i)c}{2m_2}}$$

$$D = \sqrt{c^2 - 4km_2}, H_i = \text{Heaviside}(t - t_i)$$

Eqs. (17) and (18) calculate the residual vibration response in terms of displacement and acceleration. However, the objective here is to examine the amplitude of residual vibration at the tip end after TT or Δt . Therefore, a time shift (t_c) is introduced, which is the time required after Δt to reach the

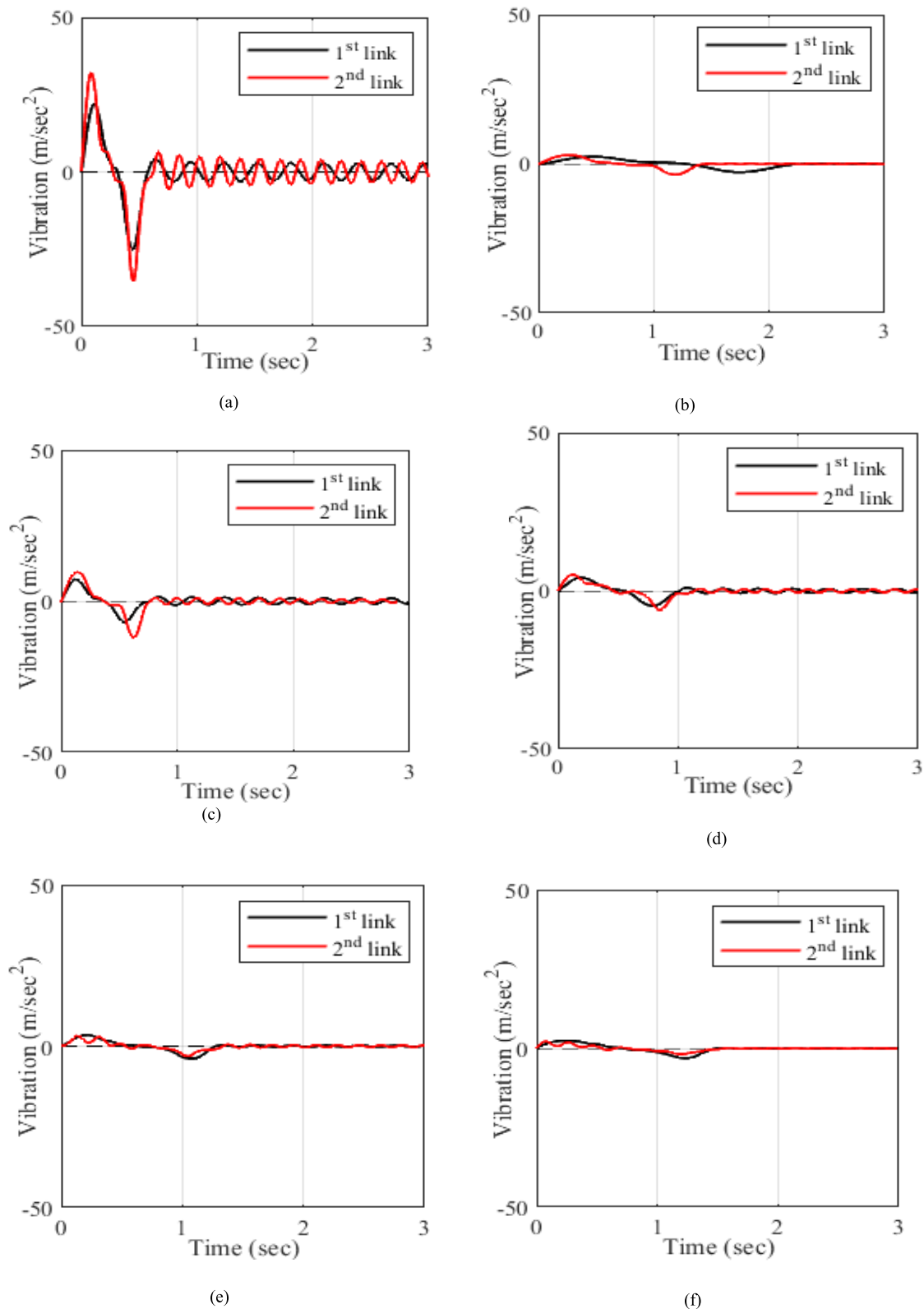


FIGURE 14. Experimentally obtained vibration response with respect to time for 1st and 2nd flexible link: (a) Conventional fourth-order S-curve; (b) Optimum fourth-order S-curve, (c) Optimum 15- SFOS S-curve for 0.75 sec TT; (d) Optimum 15- SFOS S-curve for 1 sec TT; (e) Optimum 15- SFOS S-curve for 1.25 sec TT; (f) Optimum 15- SFOS S-curve for 1.5 sec TT; (g) Optimum 15- SFOS S-curve for 1.75 sec TT; (h) Optimum 15- SFOS S-curve for 2 sec TT profiles.

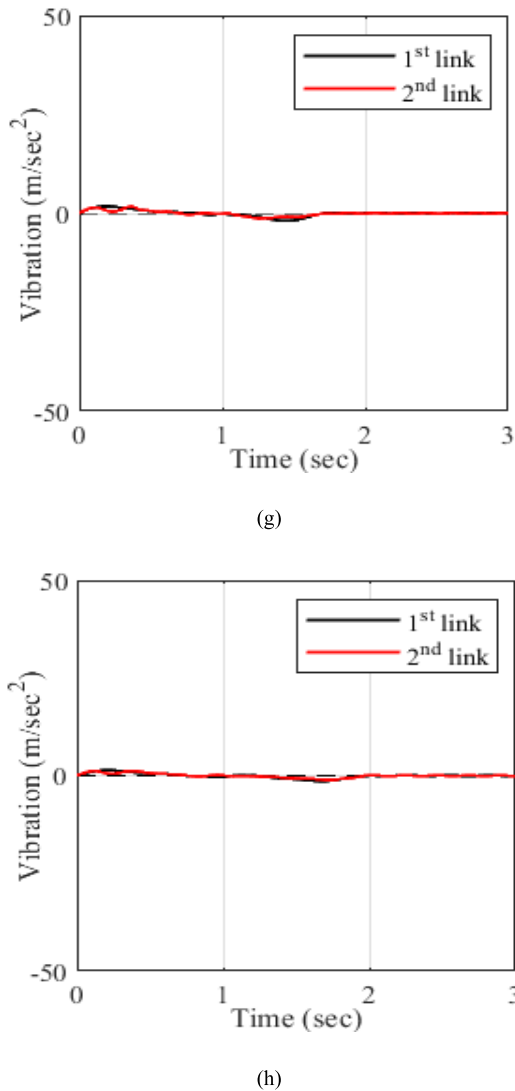


FIGURE 14. (Continued.) Experimentally obtained vibration response with respect to time for 1st and 2nd flexible link: (a) Conventional fourth-order S-curve; (b) Optimum fourth-order S-curve; (c) Optimum 15-SFOS S-curve for 0.75 sec TT; (d) Optimum 15-SFOS S-curve for 1 sec TT; (e) Optimum 15-SFOS S-curve for 1.25 sec TT; (f) Optimum 15-SFOS S-curve for 1.5 sec TT; (g) Optimum 15-SFOS S-curve for 1.75 sec TT; (h) Optimum 15-SFOS S-curve for 2 sec TT profiles.

magnitude of the first and the highest amplitude of residual vibration: $t_c = t - \Delta t$. Then ' t' ' becomes:

$$t = \Delta t + t_c \quad (19)$$

Substituting Eq. (19) into Eqs. (17) and (18), the residual vibration amplitude in terms of displacement and acceleration after Δt can be computed. The general form of the residual vibration response, in terms of acceleration at the tip end of the flexible rotating link generated using Eq. (18), is depicted in Fig. 8. The plot illustrates the response of forced vibration from 0 to Δt , which represents the response observed during the motion. The free vibration response is observed after Δt or upon completion of the motion. Additionally, this plot defines the time shift observed between the time periods of $(\Delta t + t_c)$ and Δt .

V. EXPERIMENTATION

A. EXPERIMENTAL SETUP

The experiment is conducted on a flexible rotating link to demonstrate the practical improvement achieved by using the proposed 15-SFOS S-curve profile. The experimental setup of the flexible rotating link and its schematic arrangement is shown in Fig. 9. The flexible rotating link is a thin stainless-steel beam attached perpendicular to the shaft of the DC motor using a rotating hub. This arrangement restricts the flexible link to rotate in only horizontal plane for a prescribed P-T-P angular movement. The installed DC motor comes with an encoder sensor used to measure the angular movement ($\theta(t)$). Meanwhile, an accelerometer is mounted at the tip end of the flexible rotating link to measure tip transverse vibration. The experiments are conducted with two different natural frequencies ($\omega_1 = 3.49$ Hz and $\omega_2 = 5.64$ Hz) for the flexible rotating link, each having a length of 0.296 m. Although both flexible links have equal lengths and heights (0.024 m), the variation in the natural frequency arises due to their differing widths (0.00066 m, 0.00095 m). A mass of 0.02 kg is also attached at the tip end of each flexible rotating links to further alter their natural frequencies. The parameters of the experimental setup for the flexible rotating link are listed in Table 4. The natural frequency and damping ratio are experimentally determined for each of the flexible rotating links. V_{max} , A_{max} , J_{max} , and S_{max} are determined by the motor's capacity. To execute the prescribed P-T-P motion, a microcontroller (Teensy 3.6) is employed as a signal generator, producing 12-bit PWM driving voltage signals at 15 kHz with a sampling interval of 250 μ s. These signals are regulated by the motor drive to actuate the motor according to the specified motion profile. The complete process is illustrated in the block diagram shown in Fig. 10.

For a given set of values for V_{max} , A_{max} , J_{max} , and S_{max} , the 15-SFOS S-curve profile computes the minimum TT as 0.559 sec. [42]. However, this minimum TT, resulted in a very high residual vibration amplitude (refer to Table 8). It is crucial to increase the TT of the P-T-P motion to reduce the residual vibration amplitude to a more manageable level. Therefore, to demonstrate the effectiveness of the proposed CMP method, the experiments are conducted with TT values of 0.75, 1, 1.25, 1.5, 1.75, and 2 sec.

The residual vibration response at the tip of the link, measured using an accelerometer mounted at the tip, is presented in the bottom plot of Fig. 11 for one of the experiments (1st link and minimum TT). The upper plot illustrates the comparison between the planned and measured angular velocity in rad/sec for the same motion. The angular velocity is measured using an incremental encoder. As observed in Fig. 11, the measured angular velocity agrees well with the planned angular velocity.

In Fig. 11, point 'A' denotes the amplitude of residual vibration immediately following the completion of motion. This point corresponds to the maximum residual vibration magnitude observed in the free vibration region. The residual vibration amplitude, measured in terms of acceleration

TABLE 9. Advantages of the presented method over the previously published methods.

Advantages	Presented Method	Previously Published Method
Vibration reduction capability	1) It can produce the near-optimum solution to minimize residual vibration.	[37-38], [41-43]: - 1) It can optimize the TT but does not appropriately suppress the residual vibration.
Duration of P-T-P motion	1) The TT can be controllable. 2) It can provide minimum residual vibration response for specific TT. 3) The TT can be minimized with a slight compromise in the residual vibration amplitude.	[44-46]: - 1) The TT cannot be controlled. 2) The TT is determined according to the system dynamics and its parameters.
Dynamics of system	1) It does not involve any dynamics of the system and its parameters. 2) There is no need to perform any computations; the appropriate magnitudes can directly be calculated using provided nomograms.	[44-46]: - 1) Involved complex dynamics. 2) Required to handle complex mathematics, thereby increasing the computational burden.
Practical application	1) It provides a generalized solution that can be used for flexible systems and any electromechanical P-T-P motion system.	[44-46]: - 1) It can only apply to lightly damped flexible systems. 2) Since it involves dynamics, the solution changes as the system dynamics change.

(m/s²), is determined solely based on point 'A' for each experiment. The total time to reach this point is equal to the total motion time (TT) plus a time shift, as illustrated in Fig. 8.

B. DETERMINATION OF MULTIPLICATION FACTORS AND UNITIZED LIMITING MAGNITUDES

The first step in obtaining the optimum 15 SFOS S-curve profile is to determine the MF for each MP. The MF for different TT is determined according to Table 1 and shown in Table 5. In the next step, the unitized limiting magnitudes (ULM) of velocity (ULM_v), acceleration (ULM_a), jerk (ULM_j), and snap (ULM_s) are calculated for different TT by dividing the magnitudes of given actuator capacities with their respective MF as given in Table 5. The ULM of MP for different TT is shown in Table 6. Considering these ULM, the optimum UM of MP based on critical acceleration, critical jerk, and critical snap are determined according to the flowchart shown in Fig. 6.

C. DETERMINATION OF OPTIMUM UNITIZED AND ACTUAL MAGNITUDES OF MOTION PARAMETERS

The optimum AM of the MP is determined by multiplying the optimum UM with the MF listed in Table 5. The optimum UM and AM of MP, as per the CMP, for different TT values, are provided in Table 7. Furthermore, using these optimum AM, the optimum 15-SFOS S-curve profiles are generated

for varying TT. The experiments are conducted for each of the optimum profiles. The global optimum 15-SFOS S-curve profile will be the one that results in the smallest residual vibration amplitude.

The experimentally determined residual vibration amplitude at the end of the motion, corresponding to each of the optimum 15-SFOS S-curve profiles, are illustrated in Fig. 12 for two different flexibilities of the links and various TT. It is observed that, for TT ranging from 0.75 to 1 sec, the profile with critical acceleration and critical jerk results in lower residual vibration amplitudes for the first and second links, respectively. For TT ranging from 1.25 to 2 sec, a profile with a critical jerk and critical snap results in lower residual vibration amplitudes for the first and second links, respectively. The profile with critical snap does not yield a global optimum solution for the first link. Fig. 13 shows the minimum residual vibration amplitudes for both links as determined through experimentation and simulation. The comparison reveals that the experimental results agree well with the simulated results.

D. PERFORMANCE COMPARISON OF PRESENTED METHOD WITH THE TRADITIONAL METHOD

The efficacy of the proposed method is demonstrated through a comparison of the performance of other traditional fourth-order trajectories. The time parameters of these trajectories are determined using the parameters listed in Table 4. A quantitative comparative analysis is conducted among these trajectory models based on three parameters: namely, the TT required to complete the P-T-P motion, the ability to reduce residual vibration amplitude at the end of the P-T-P motion, and the requirement of system parameters as detailed in Table 8. The plots in Fig. 14 depict the residual vibration response of the flexible link for various trajectory models, which is measured by the accelerometer attached to the end of the flexible link. The black line represents the response of the 1st link, while the red line represents the response of the 2nd link.

The referred traditional fourth-order S-curve profile is utilized in various industrial applications of electromechanical systems. Therefore, it is commonly employed in most of the literature for comparison, showcasing the effectiveness of the respective works. In this context, alongside traditional optimum fourth-order S-curve profile, traditional fourth-order S-curve profile is also used for comparison, demonstrating the feasibility of the proposed 15-SFOS S-curve trajectory. The formulation and algorithm necessary to generate traditional fourth-order S-curve profile can be found in reference [32]. The methods used to create various trajectories are illustrated in the third column of Table 8. The traditional fourth-order S-curve profile is essentially generated using a simple time-optimal trajectory planning method, in which actuator capacities play a crucial role in profile design. Meanwhile, the fourth-order S-curve profile is created using a general optimization method defined in the Laplace domain.

The traditional optimum fourth-order S-curve profile requires two additional initial system parameters, namely natural frequency and damping ratio, due to the involvement of system dynamics in designing the appropriate profile. In contrast, the presented trajectory does not require any initial system parameters as it does not involve system dynamics in designing the appropriate profile. This represents a major advantage of the proposed trajectory model over the traditional optimum fourth-order S-curve profile. Since the presented work focuses on minimizing the residual vibration for any specific duration of the P-T-P motion, it requires one extra initial parameter, namely TT. This common objective and the equitable initial situation between the presented work and the works considered for comparison create a fair competitive condition, which is crucial when comparing results between two methods.

As observed in Table 8, when compared to the traditional fourth-order S-curve profile, the referred optimum profile and the presented optimum profile can significantly suppress induced residual vibration. Although the TT for the traditional fourth-order S-curve profile is very short, and it does not require the system parameters, it is incapable of producing low residual vibration amplitude. The presented profile significantly suppresses induced residual vibration when compared to traditional fourth-order S-curve profile. Specifically, at a TT of 2 sec., the presented profile reduces residual vibration by 98.3% for the 1st link, and by 99.05% for the 2nd link, compared to the traditional fourth-order S-curve profile. The presented profile performs similarly to the traditional optimum fourth-order S-curve profile. Compared to the traditional optimum fourth-order S-curve profile, the presented S-curve profile, at a TT of 2 sec., increases residual vibration by only 36.5 % for the 1st link, while reducing residual vibration by approximately 56.3% for the 2nd link.

The presented profile performs at a similar level to that of the traditional optimum fourth-order S-curve profile. However, the TT of the traditional optimum fourth-order S-curve profile cannot be controlled, as it is calculated based on the system parameters. If there's a desire to achieve a lower TT with a slight compromise in the amplitude of residual vibration, the traditional optimum fourth-order S-curve profile cannot offer a solution. On the other hand, the presented profile can control the TT, thereby providing the solution for attaining a lower TT with reduced residual vibration response. When comparing the results of the presented profile for the 1st link with a TT of 1.75 sec to the results of the 1st link with the traditional optimum fourth-order S-curve profile, an increase of 52.7% in the amplitude of residual vibration is observed. This increase can be acceptable since it occurs at a lower amplitude level of residual vibration. In contrast, the TT of the optimum fourth-order S-curve profile is 2.292 sec., which is 23.6% higher than the 1.75 seconds TT of the presented profile. This increase in TT is noticeable since it is observed at the higher level of TT. Therefore, if there's a need to achieve a lower TT with a minimum amplitude

of residual vibration, the presented method, can be useful in providing the desired solution.

Furthermore, traditional model-based approaches require a detailed understanding of the system dynamics and associated parameters, and they involve solving complex dynamic equations. This leads to significant mathematical effort and increases the computational burden when designing the motion profile. In contrast, the proposed model does not require any knowledge of the system parameters or the solution of dynamic equations. The optimal profile can be obtained simply by following the flowchart presented in Fig. 6, thereby eliminating the need for intricate mathematics and reducing computational effort. Consequently, the presented model can more effectively suppress the amplitude of induced residual vibration with a shorter TT compared to the traditional models considered in the comparison. Table 9 provides a concise summary of the advantages of the presented method in contrast to the various methods employed in trajectory planning.

VI. CONCLUSION

This work provides a simple and effective way to control the induced residual vibration of the flexible system. In this paper, the new concept of CMP is introduced to obtain the near optimum 15-SFOS S-curve profile for achieving the low-vibration P-T-P motion, for a given moving distance, TT, and actuator capacities, without explicitly addressing the system's dynamics. Based on CMP, the magnitudes of NCMP are obtained using the presented nomograms. A flowchart is also provided, in which the procedure to determine the optimum magnitudes of MP based on the CMP is depicted. The feasibility of the proposed method is experimentally demonstrated on the setup of flexible rotating links. The experimental results are also validated through simulated results.

The following important conclusions can be drawn based on this work:

- 1) The CMP method presented in this paper optimizes the 15-SFOS S-curve profile to minimize residual vibration for specific TT and displacement within the actuator capacities.
- 2) The method presented does not rely on system dynamics and their parameters to achieve low-vibration P-T-P motion.
- 3) The novel concept of unitization introduced in this paper enables the generalization of designing a 15-SFOS S-curve profile for a given TT and displacement.
- 4) The method presented allows for the optimization of the 15-SFOS S-curve profile using any of the three CMP: critical acceleration, critical jerk, and critical snap.
- 5) This method provides a generalized solution applicable to any electromechanical P-T-P motion system for minimizing residual vibration.
- 6) The plots in Figs. 2 and 3 are suitable for qualitative manipulation of MP magnitudes and are valuable for profile planning to minimize residual vibration. They

can serve as nomograms for designing third and fourth-order S-curve profiles.

Future studies will concentrate on developing the optimum solution for different magnitudes of CMP, which enhances the capability of suppressing the residual vibration. Finding the precise CMP will also be a focus of future work to minimize the number of experiments needed.

Conflict of Interest

The authors declare no competing interests.

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